STAT 339 METHODS OF LINEAR ALGEBRA
Determinants

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Outline

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The Determinant of a Matrix

The determinant of a 2 × 2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\implies det(A) = |A| = a_{11}a_{22} - a_{12}a_{21}$$

Note

1. For every SQUARE matrix, there is a real number associated with this matrix and called its determinant
2. It is common practice to omit the matrix brackets

$$\left| \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

The Determinant of a Matrix

The determinant of a 2 × 2 matrix

Historically speaking, the use of determinants arose from the recognition of special patterns that occur in the solutions of linear systems

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

$$\implies x_1 = \frac{b_1 a_{22} - b_2 a_{12}}{a_{11}a_{22} - a_{12}a_{21}} \text{ and } x_2 = \frac{b_2 a_{11} - b_1 a_{21}}{a_{11}a_{22} - a_{12}a_{21}}$$

Note

1. x_1 and x_2 have the same denominator, and this quantity is called the determinant of the coefficient matrix A.
2. There is a unique solution if $a_{11}a_{22} - a_{12}a_{21} = |A| \neq 0$

The Determinant of a Matrix

Example (The determinant of a 2×2 matrix)

$$\begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(-3) = 4 + 3 = 7$$

$$\begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 2(2) - 1(4) = 4 - 4 = 0$$

$$\begin{vmatrix} 0 & 3/2 \\ 2 & 4 \end{vmatrix} = 0(4) - 2(3/2) = 0 - 3 = -3$$

The Determinant of a Matrix

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$$\begin{vmatrix} 0 & 3/2 \\ 2 & 4 \end{vmatrix} = 0(4) - 2(3/2) = 0 - 3 = -3$$

Note

1. The determinant of a matrix can be positive, zero, or negative

The Determinant of a Matrix

The determinant of a $n \times n$ matrix

- Minor of the entry a_{ij}: the determinant of the matrix obtained by deleting the i^{th} row and j^{th} column of A

$$M_{ij} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1(j-1)} & a_{1(j+1)} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ a_{(i-1)1} & a_{(i-1)2} & \cdots & a_{(i-1)(j-1)} & a_{(i-1)(j+1)} & \cdots & a_{(i-1)n} \\ a_{(i+1)1} & a_{(i+1)2} & \cdots & a_{(i+1)(j-1)} & a_{(i+1)(j+1)} & \cdots & a_{(i+1)n} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n(j-1)} & a_{n(j+1)} & \cdots & a_{nn} \end{bmatrix}$$

  M_{ij} is a real number

- Cofactor of a_{ij}

$$C_{ij} = (-1)^{i+j}|M_{ij}|$$

  C_{ij} is also a real number

Example (The determinant of a square matrix of order 3)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\implies M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} \qquad M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{13} & a_{33} \end{vmatrix}$$

$$\implies C_{21} = (-1)^{2+1} M_{21} = -M_{21} \qquad C_{22} = (-1)^{2+2} M_{22} = M_{22}$$

The Determinant of a Matrix

Example (The determinant of a square matrix of order 3)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

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$$\implies C_{21} = (-1)^{2+1} M_{21} = -M_{21} \qquad C_{22} = (-1)^{2+2} M_{22} = M_{22}$$

Note

1. Sign pattern for cofactors. Odd positions (where $i+j$ is odd) have negative signs, and even positions (where $i+j$ is even) have positive signs. (Positive and negative signs appear alternately at neighboring positions.)

The Determinant of a Matrix

Theorem ((3.1) Expansion by cofactors)

Let A be a square matrix of order n, then the determinant of A is given by

1. $det(A) = |A| = \sum_{j=1}^{n} a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$, *(cofactor expansion along the $i-th$ row, $i = 1, 2, \ldots, n$)*
2. $det(A) = |A| = \sum_{i=1}^{n} a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$, *(cofactor expansion along the $j-th$ row, $j = 1, 2, \ldots, n$)*

Note

1. The determinant can be derived by performing the cofactor expansion along any row or column of the examined matrix

Example (The determinant of a square matrix of order 3)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{aligned}
\implies \det(A) =& a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \text{(first row expansion)} \\
=& a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} \text{(second row expansion)} \\
=& a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \text{(third row expansion)} \\
=& a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} \text{(first column expansion)} \\
=& a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} \text{(second column expansion)} \\
=& a_{13}C_{31} + a_{23}C_{23} + a_{33}C_{33} \text{(third column expansion)}
\end{aligned}$$

Note

1. The determinant can be derived by performing the cofactor expansion along any row or

The Determinant of a Matrix

Example (The determinant of a square matrix of order 3)

Find the determinate of A

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

Solution (The determinant of a square matrix of order 3)

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$\implies det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} (\textit{first row expansion})$$

$$But\ C_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \quad C_{12} = (-1)^{1+2} \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} = 5 \quad C_{13} = (-1)^{1+3} \begin{vmatrix} 3 & -1 \\ 4 & 0 \end{vmatrix} = 4$$

$$\therefore det(A) = (0)(1) + (2)(5) + 1(4) (\textit{first row expansion})$$
$$= 14$$

The Determinant of a Matrix

Example (Alternative way to calculate the determinant of a square matrix of order 3)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{aligned} \implies \det(A) =& a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ & - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12} \end{aligned}$$

How to arrived at the above expression will be demonstrated in class

Note

1. This method is only valid for matrices with the order of 3

The Determinant of a Matrix

Example (Alternative way to calculate the determinant of a square matrix of order 3)

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \implies \det(A) =&(0)(-1)(1) + (2)(2)(4) + (1)(3)(0) \\ &-(4)(-1)(1) - (0)(2)(0) - (1)(3)(2) \\ =&0 + 16 + 0 - (-4) - 6 = 14 \end{aligned}$$

Example (The determinant of a square matrix of order 4)

Find the determinate of A

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}$$

The Determinant of a Matrix

Solution (The determinant of a square matrix of order 4)

$$
\begin{aligned}
det(A) =& (3)C_{13} + (0)C_{23} + (0)C_{33} + (0)C_{43} = (3)C_{13} \\
=& (3)(-1)^{1+3} \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} \\
=& 3\left[(0)(-1)^{2+1}\begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} + (2)(-1)^{2+2}\begin{vmatrix} -1 & 2 \\ 3 & 2 \end{vmatrix} + (3)(-1)^{2+3}\begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix}\right] \\
=& 3[0 + 2(1)(-4) + (3)(-1)(-7)] = 3(13) = 39
\end{aligned}
$$

Note

1. By comparing this example with the previous, it is apparent that the computational effort for the determinant of 4×4 matrices is much higher than that of 3×3 matrices. In the next section, we will learn a more efficient way to calculate the determinant.

The Determinant of a Matrix

Theorem ((3.2) Determinant of a Triangular Matrix)

If A is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then its determinant is the product of the entries on the main diagonal. That is

$$det(A) = |A| = \prod_{i=1}^{n} a_{ii} = a_{11} \cdot a_{22} \cdot \cdots \cdot a_{nn}$$

Note

1. Based on this Theorem 3.2, it is straightforward to obtain that det(I)=1.
2. On the next slide, the case of upper triangular matrices will be used to prove Theorem 3.2. It is straightforward to apply the following proof for the cases of lower triangular and diagonal matrices.

The Determinant of a Matrix

By Mathematical Induction.

Suppose that the theorem is true for any upper triangular matrix U of order $n-1$, that is,

$$|U| = a_{11}a_{22}\cdots a_{(n-1)(n-1)}$$

The Determinant of a Matrix

By Mathematical Induction.

Suppose that the theorem is true for any upper triangular matrix U of order $n-1$, that is,

$$|U| = a_{11}a_{22}\cdots a_{(n-1)(n-1)}$$

Then consider the determinant of an upper triangular matrix A of order n by the cofactor expansion across the $n-$th row

$$|A| = (0)C_{n1} + (0)C_{n2} + \cdots + (a_{nn})C_{nn} = (a_{nn})(-1)^{n+n}M_{nn} = a_{nn}M_{nn}$$

The Determinant of a Matrix

By Mathematical Induction.

Suppose that the theorem is true for any upper triangular matrix U of order $n-1$, that is,

$$|U| = a_{11}a_{22}\cdots a_{(n-1)(n-1)}$$

Then consider the determinant of an upper triangular matrix A of order n by the cofactor expansion across the $n-$th row

$$|A| = (0)C_{n1} + (0)C_{n2} + \cdots + (a_{nn})C_{nn} = (a_{nn})(-1)^{n+n}M_{nn} = a_{nn}M_{nn}$$

Since M_{nn} is the determinant of a $(n-1)\times(n-1)$ upper triangular matrix by deleting the n-th row and $n-$th column of A, we can apply the induction assumption to write

$$|A| = a_{nn}M_{nn} = a_{nn}(a_{11}a_{22}\cdots a_{(n-1)(n-1)}) = a_{11}a_{22}\cdots a_{(n-1)(n-1)}a_{nn}$$

□

The Determinant of a Matrix

Example (Determinant of a Triangular Matrix)

Find the determinants of the following triangular matrices

$$(a) A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{bmatrix} \qquad (b)\ B = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{bmatrix}$$

Evaluation of a Determinant Using Elementary Row Operations

- The computational effort to calculate the determinant of a square matrix with a large number of n is unacceptable. In this section, I will show how to reduce the computational effort by using elementary operations

Evaluation of a Determinant Using Elementary Row Operations

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Theorem ((3.3) Elementary row operations and determinants)

Let A and B be square matrices

1. $B = I_{ij}(A) \implies det(B) = -det(A)$
2. $B = M_i^{(k)}(A) \implies det(B) = k\,det(A)$
3. $B = A_{ij}^{(k)}(A) \implies det(B) = det(A)$

Evaluation of a Determinant Using Elementary Row Operations

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3. $B = A_{ij}^{(k)}(A) \implies det(B) = det(A)$

Note

1. The above three properties remains valid if elementary column operations are performed to derive column-equivalent matrices

Evaluation of a Determinant Using Elementary Row Operations

Example (Three Properties of Elementary row operations and determinants)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 2 & 1 \end{bmatrix} \implies \det(A) = -2$$

- (P1)

$$A_1 = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} \implies \det(A_1) = 2$$

$$\begin{aligned} A_1 &= I_{1,2}(A) \\ \det(A_1) &= -\det(A) = -(-2) = 2 \end{aligned}$$

Evaluation of a Determinant Using Elementary Row Operations

Example (Three Properties of Elementary row operations and determinants)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 2 & 1 \end{bmatrix} \implies \det(A) = -2$$

- (P2)

$$A_2 = \begin{bmatrix} 4 & 8 & 12 \\ 0 & 1 & 4 \\ 1 & 2 & 1 \end{bmatrix} \implies \det(A_2) = -8$$

$$\begin{aligned} A_2 &= M_1^{(4)}(A) \\ \det(A_2) &= 4\det(A) = 4(-2) = -8 \end{aligned}$$

Example (Three Properties of Elementary row operations and determinants)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 2 & 1 \end{bmatrix} \implies \det(A) = -2$$

- (P3)

$$A_3 = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -3 & -2 \\ 1 & 2 & 1 \end{bmatrix} \implies \det(A_3) = -2$$

$$\begin{aligned} A_3 &= A_{1,2}^{(-2)}(A) \\ \det(A_3) &= \det(A) = -2 \end{aligned}$$

Row reduction method to evaluate the determinant

- A row-echelon form of a square matrix is either an upper triangular matrix or a matrix with zero rows.
- It is easy to calculate the determinant of an upper triangular matrix (by Theorem) or a matrix with zero rows ($\det = 0$).

Example (Evaluation a determinant using elementary row operations)

$$A = \begin{bmatrix} 2 & -3 & 10 \\ 1 & 2 & -2 \\ 0 & 1 & -3 \end{bmatrix} \implies \det(A) = ?$$

$$\det(A) = \begin{vmatrix} 2 & -3 & 10 \\ 1 & 2 & -2 \\ 0 & 1 & -3 \end{vmatrix} \xrightarrow{I_{1,2}} -\overbrace{\begin{vmatrix} 1 & 2 & -2 \\ 2 & -3 & 10 \\ 0 & 1 & -3 \end{vmatrix}}^{A_1} \xrightarrow{A_{1,2}^{(-2)}} -\overbrace{\begin{vmatrix} 1 & 2 & -2 \\ 0 & -7 & 14 \\ 0 & 1 & -3 \end{vmatrix}}^{A_2} \xrightarrow{M_2^{(-1/7)}} (-1)(\frac{1}{-1/7})\overbrace{\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -3 \end{vmatrix}}^{A_3}$$

$$\xrightarrow{A_{2,3}^{(-1)}} 7\overbrace{\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -1 \end{vmatrix}}^{A_4} = 7(-1) = 7$$

Evaluation of a Determinant Using Elementary Row Operations

Example (Evaluation a determinant using elementary row operations)

$$A = \begin{bmatrix} 2 & 3 & 10 \\ 1 & 2 & -2 \\ 0 & 1 & -3 \end{bmatrix} \implies \det(A) = ?$$

Note

$$\begin{aligned}
\det(A_1) &= -\det(A) \implies \det(A) = -\det(A_1) \\
\det(A_2) &= \det(A_1) \\
\det(A_3) &= -\frac{1}{7}\det(A_2) \implies \det(A_2) = \frac{1}{-1/7}\det(A_3) \\
\det(A_4) &= \det(A_3)
\end{aligned}$$

Evaluation of a Determinant Using Elementary Row Operations

Comparison between the number of required operations for the two kinds of methods to calculate the determinant

	Co-factor Expansion		Row Reduction	
Order n	Additions	Multiplications	Additions	Multiplications
3	5	9	5	10
5	119	205	30	45
10	3,628,799	6,235,300	285	339

Note

When evaluating a determinant by hand, you can sometimes save steps by integrating this two kinds of methods (see Examples on the next three slides)

Evaluation of a Determinant Using Elementary Row Operations

Example

- Evaluating a determinant using column reduction and cofactor expansion.

$$A = \begin{bmatrix} -3 & 5 & 2 \\ 2 & -4 & -1 \\ -3 & 0 & 6 \end{bmatrix}$$

Evaluating a determinant using column reduction and cofactor expansion

Solution

$$det(A) = \begin{vmatrix} -3 & 5 & 2 \\ 2 & -4 & -1 \\ -3 & 0 & 6 \end{vmatrix} \overset{AC_{1,3}^{(2)}}{=} \begin{vmatrix} -3 & 5 & -4 \\ 2 & -4 & 3 \\ -3 & 0 & 0 \end{vmatrix} = (-3)(-1)^{3+1} \begin{vmatrix} 5 & -4 \\ -4 & 3 \end{vmatrix}$$

$$=(-3)(1)(-1) = 3$$

Note

$AC_{i,j}^{(k)}$ is the counterpart column operation to the row operation $A_{i,j}^{(k)}$

Evaluating a determinant using column reduction and cofactor expansion

Example

- Evaluating a determinant using both row and column reductions and cofactor expansion.

$$\begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix}$$

Evaluating a determinant using column reduction and cofactor expansion

Solution

$$det(A) = \begin{vmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 & 0 \end{vmatrix} \overset{A_{2,4}^{(1)} A_{2,5}^{(-1)}}{=} \begin{vmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 1 & 0 & 5 & 6 & -4 \\ 3 & 0 & 0 & 0 & 1 \end{vmatrix} = (1)(-1)^{2+2} \begin{vmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 1 & 5 & 6 & -4 \\ 3 & 0 & 0 & 1 \end{vmatrix}$$

$$\overset{AC_{4,1}^{(-3)}}{=} \begin{vmatrix} 8 & 1 & 3 & -2 \\ -8 & -1 & 2 & 3 \\ 13 & 5 & 6 & -4 \\ 0 & 0 & 0 & 1 \end{vmatrix} = (1)(-1)^{4+4} \begin{vmatrix} 8 & 1 & 3 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix} \overset{A_{2,1}^{(1)}}{=} \begin{vmatrix} 0 & 0 & 5 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix}$$

$$=(5)(-1)^{1+3} \begin{vmatrix} -8 & -1 \\ 13 & 5 \end{vmatrix} = (5)(-27) = -135$$

Evaluating a determinant using row/column reduction and cofactor expansion

Theorem ((3.4) Conditions that yield a zero determinant)

If A is a square matrix and any of the following conditions is true, then $\det(A) = 0$

1. *An entire row (or an entire column) consists of zeros*
   (Perform the cofactor expansion along the zero row or column)
2. *Two rows (or two columns) are equal*
3. *One row (or column) is a multiple of another row (or column)*
   (For (2) and (3), based on the mathematical induction, perform the cofactor expansion along any row or column other than these two rows or columns)

Note

For conditions (2) or (3), you can also use elementary row or column operations to create an entire row or column of zeros and obtain the results by the Theorem 3.3.

Evaluating a determinant using column reduction and cofactor expansion

Note

Thus, we can conclude that a square matrix has a determinant of zero if and only if it is row- (or column-) equivalent to a matrix that has at least one row (or column) consisting entirely of zeros

Example

- Evaluating a determinant using both row and column reductions and cofactor expansion.

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 4 & 5 & 6 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 4 & 2 \\ 1 & 5 & 2 \\ 1 & 6 & 2 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ -2 & -4 & -6 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 8 & 4 \\ 2 & 10 & 5 \\ 3 & 12 & 6 \end{vmatrix} = 0$$

Properties of Determinants

Theorem ((3.5) Determinant of a matrix product)

$$det(AB) = det(A)det(B)$$

Note

1. $\det(A_1 A_2 \cdots A_n) = \det(A_1)\det(A_2)\cdots\det(A_n)$
2. $$\begin{cases} \det(E_{i,j}A) = \det(E_{i,j})\det(A) \text{ and } \det(E_{i,j}A) = -\det(A) \implies \det(E_{i,j}) = -1 \\ \det(E_i^{(k)}A) = \det(E_i^{(k)})\det(A) \text{ and } \det(E_i^{(k)}A) = k\det(A) \implies \det(E_i^{(k)}) = k \\ \det(E_{i,j}^{(k)}A) = \det(E_{i,j}^{(k)})\det(A) \text{ and } \det(E_{i,j}^{(k)}A) = \det(A) \implies \det(E_{i,j}^{(k)}) = 1 \end{cases}$$
3. $\det(A + B) \neq \det(A) + \det(B)$

Properties of Determinants

Theorem (Determinant of a matrix product)

$$det(AB) = det(A)det(B)$$

Note

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$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix}$$

(There is an example to verify this property on Slides 40 & 41) (Note that this property is also valid for any rows or columns other than the second row)

Properties of Determinants

Example (The determinant of a matrix product)

- Find $|A|$, $|B|$, and $|AB|$.

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix}$$

Solution (The determinant of a matrix product)

$$|A| = \begin{vmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{vmatrix} = -7 \qquad |B| = \begin{vmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{vmatrix} = 11$$

$$AB = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{vmatrix} = -77$$

Check: $|AB| = |A||B|$

Properties of Determinants

Example (The determinant of a matrix product)

- Show that.

$$A = \begin{vmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{vmatrix} \stackrel{?}{=} \begin{vmatrix} 1 & -2 & 2 \\ -1 & 1 & 2 \\ 1 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -2 & 2 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix} = |B| + |C|$$

Properties of Determinants

Proof.

$$|A| = 0(-1)^{2+1}\begin{vmatrix}-2 & 2\\ 0 & 1\end{vmatrix} + 3(-1)^{2+2}\begin{vmatrix}1 & 2\\ 1 & 1\end{vmatrix} + 2(-1)^{2+3}\begin{vmatrix}1 & -2\\ 1 & 0\end{vmatrix}$$

$$|B| = -1(-1)^{2+1}\begin{vmatrix}-2 & 2\\ 0 & 1\end{vmatrix} + 1(-1)^{2+2}\begin{vmatrix}1 & 2\\ 1 & 1\end{vmatrix} + 2(-1)^{2+3}\begin{vmatrix}1 & -2\\ 1 & 0\end{vmatrix}$$

$$|C| = 1(-1)^{2+1}\begin{vmatrix}-2 & 2\\ 0 & 1\end{vmatrix} + 2(-1)^{2+2}\begin{vmatrix}1 & 2\\ 1 & 1\end{vmatrix} + 0(-1)^{2+3}\begin{vmatrix}1 & -2\\ 1 & 0\end{vmatrix}$$

$\square$

Theorem ((3.6) Determinant of a scalar multiple of a matrix)

If A is an $n \times n$ matrix and c is a scalar, then

$$det(cA) = c^n det(A)$$

(can be proven by repeatedly use the fact that if $B = M_i^{(k)}(A) \implies |B| = k|A|$)

Example

$$A = \begin{bmatrix} 10 & -20 & 40 \\ 30 & 0 & 50 \\ -20 & -30 & 10 \end{bmatrix}, \text{ if } \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = 5, \text{ find } |A|$$

Properties of Determinants

Solution (Determinant of a scalar multiple of a matrix)

$$A = 10\begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{bmatrix} \implies |A| = 10^3\begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = (1000)(5) = 5000$$

Properties of Determinants

Theorem ((3.7) Determinant of an invertible matrix)

A square matrix A is invertible (nonsingular) if and only if $\det(A) \neq 0$.

Proof.

($\Longrightarrow$) If A is invertible, then $AA^{-1} = I$. By Theorem 3.5, we can have $|A||A^{-1}| = |I|$. Since $|I| = 1$, neither $|A|$ nor $|A^{-1}|$ is zero

Suppose $|A|$ is nonzero. It is aimed to prove A is invertible. By the Gauss-Jordan elimination, we can always find a matrix B, in reduced row-echelon form, that is row-equivalent to A

1. Either B has at least one row with entire zeros, then $|B| = 0$ and thus $|A| = 0$ since $|E_k| \cdots |E_2||E_1||A| = |B|$.
2. Or $B = I$, then A is row-equivalent to I, and by previous Theorem, it can be concluded that A is invertible.

□

Example

Classifying square matrices as singular or nonsingular

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ -3 & -2 & -1 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ -3 & -2 & 1 \end{bmatrix} = 5$$

Solution

$|A| = 0 \implies$ *A has no inverse (it is singular)*

$|B| = -12 = \neq 0 \implies$ *B has inverse (it is nonsingular)*

Properties of Determinants

Theorem ((3.8) Determinant of an inverse matrix)

If A is invertible (nonsingular), then $det(A^{-1}) = \frac{1}{det(A)}$.

Properties of Determinants

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(Since $AA^{-1} = 1$, then $|A||A^{-1}| = 1$)

Theorem ((3.9) Determinant of an inverse matrix)

If A is a square matrix, then $det(A^T) = det(A)$.

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Properties of Determinants

Theorem ((3.8) Determinant of an inverse matrix)

If A is invertible (nonsingular), then $det(A^{-1}) = \frac{1}{det(A)}$.
(Since $AA^{-1} = 1$, then $|A||A^{-1}| = 1$)

Theorem ((3.9) Determinant of an inverse matrix)

If A is a square matrix, then $det(A^T) = det(A)$.
(Based on the mathematical induction, compare the cofactor expansion along a row of A and the cofactor expansion along the corresponding column of A^T)

Properties of Determinants

The similarity between the noninvertible matrix and the real number 0

	Matrix A	Real number c
Invertible	$\det(A) \neq 0$	$c \neq 0$
	A^{-1} exist and $\det(A^{-1}) = \frac{1}{\det(A)}$	c^{-1} exist and $c^{-1} = \frac{1}{c}$
Noninvertible	$\det(A) = 0$	$c = 0$
	A^{-1} does not exist $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{0}$	c^{-1} does not exist $c^{-1} = \frac{1}{c} = \frac{1}{0}$

Properties of Determinants

Equivalent conditions for a nonsingular matrix

IfA is an $n \times n$ matrix, then the following statements are equivalent

1. A is invertible
2. $Ax = b$ has a unique solution for every $n \times 1$ matrix b
3. $Ax = 0$ has only the trivial solution
4. A is row-equivalent to I_n
5. A can be written as the product of elementary matrices
6. $\det(A) \neq 0$

Properties of Determinants

Example

Classifying square matrices as singular or nonsingular

$$
\begin{array}{rrrrrrrr}
 & & & 2x_2 & - & x_3 & = & -1 \\
(a)A = & 3x_1 & + & 2x_2 & + & x_3 & = & 4 \\
 & 3x_1 & + & 2x_2 & - & x_3 & = & -4
\end{array}
$$

$$
\begin{array}{rrrrrrrr}
 & & & 2x_2 & - & x_3 & = & -1 \\
(b)B = & 3x_1 & + & 2x_2 & + & x_3 & = & 4 \\
 & 3x_1 & + & 2x_2 & + & x_3 & = & -4
\end{array}
$$

Applications of Determinants

- Matrix of cofactors of A

$$[C_{ij}] = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \cdots & \cdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}, \text{where } C_{ij} = (-1)^{i+j} M_{ij}.$$

- Adjoint matrix of A

$$\text{adj}(A) = [C_{ij}]^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \cdots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

Theorem ((3.10) The inverse of a matrix expressed by its adjoint matrix)

If A is an $n \times n$ invertible matrix, then

$$A^{-1} = \frac{1}{det(A)} adj(A)$$

Proof.

Consider the product $A[\text{adj}(A)]$

$$= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \cdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} C_{11} & \cdots & C_{j1} & \cdots & C_{n1} \\ C_{12} & \cdots & C_{j2} & \cdots & C_{n2} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ C_{1n} & \cdots & C_{jn} & \cdots & C_{nn} \end{bmatrix} = \begin{bmatrix} \det(A) & 0 & \cdots & 0 \\ 0 & \det(A) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \det(A) \end{bmatrix}$$

Proof.

The entry at the position (i, j) of $A[\text{adj}(A)]$

$$a_{i1}C_{j1} + a_{i2}C_{j2} + \cdots + a_{in}C_{jn} = \begin{cases} \det(A) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Consider a matrix B similar to matrix A except that the j-th row is replaced by the i-th row of matrix A

$$\implies \det(B) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 0$$

Proof.

Perform the cofactor expansion along the j-th row of matrix B

$$\implies \det(B) = a_{i1}C_{j1} + a_{i2}C_{j2} + \cdots + a_{in}C_{jn} = 0$$

(Note that $C_{j1}, C_{j2}, \cdots$ and C_{jn} are still the cofactors for the entries of the j-th row)

$$\implies A[\text{adj}(A)] = \det(A)I \implies A[\overbrace{\frac{1}{\det(A)}\text{adj}(A)}^{A^{-1}}] = I$$

$\square$

Applications of Determinants

Example (For any 2×2 matrix, its inverse can be calculated as follows)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \det(A) = ad - bc,$$

$$\operatorname{adj}(A) = \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\implies A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

$$= \frac{1}{ad - bac} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example

Given

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

1. Find the adjoint matrix of A.
2. Use the adjoint matrix of A to find A^{-1}.

Example

Given

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

1. Find the adjoint matrix of A.
2. Use the adjoint matrix of A to find A^{-1}.

Solution

1. *Find the adjoint matrix of A.*

$$\because C_{ij} = (-1)^{i+j} M_{ij}$$

Solution

1. *Find the adjoint matrix of A.*

$$C_{11} = + \begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} = 4, \quad C_{12} = - \begin{vmatrix} 0 & 1 \\ 1 & -2 \end{vmatrix} = 1, \quad C_{13} = + \begin{vmatrix} 0 & -2 \\ 1 & 0 \end{vmatrix} = 2$$

$$C_{21} = - \begin{vmatrix} 3 & 2 \\ 0 & -2 \end{vmatrix} = 6, \quad C_{22} = + \begin{vmatrix} -1 & 2 \\ 1 & -2 \end{vmatrix} = 0, \quad C_{23} = + \begin{vmatrix} -1 & 3 \\ 1 & 0 \end{vmatrix} = 3$$

$$C_{31} = + \begin{vmatrix} 3 & 2 \\ -2 & 1 \end{vmatrix} = 7, \quad C_{32} = - \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = 1, \quad C_{33} = + \begin{vmatrix} -1 & 3 \\ 0 & -2 \end{vmatrix} = 2$$

$\Longrightarrow$ *cofactors of matrix of A is*

$$[C_{ij}] = \begin{bmatrix} 4 & 1 & 2 \\ 6 & 0 & 3 \\ 7 & 1 & 2 \end{bmatrix}$$

Applications of Determinants

Solution

1. *Find the adjoint matrix of A.*

   $\Longrightarrow$ *adjoint of matrix of A is*

$$adj(A) = [C_{ij}]^T = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

2. *Use the adjoint matrix of A to find A^{-1}.*

   $\Longrightarrow$ *inverse of matrix of A is*

$$\begin{aligned} A^{-1} &= \frac{1}{det(A)} adj(A) \\ &= \frac{1}{3}\begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 4/3 & 2 & 7/3 \\ 1/3 & 0 & 1/3 \\ 2/3 & 1 & 2/3 \end{bmatrix} \end{aligned}$$

Theorem ((3.11) Cramer's Rule)

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{nn}x_n & = & b_2 \\ & & & & \vdots & & & & \\ a_{n1}x_1 & + & a_{n2}x_2 & + & \cdots & + & a_{nn}x_n & = & b_n \end{array} \implies A\mathbf{X} = \mathbf{b}$$

where

$$A = [a_{ij}] = [A^{(1)} A^{(2)} \cdots A^{(n)}], \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

Theorem (Cramer's Rule)

Suppose he system has a unique solution, that is,

$$det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{nn} \\ \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \neq 0$$

By definition,

$$\begin{aligned} A_j = [a_{ij}] =& [A^{(1)} A^{(2)} \cdots A^{(j-1)} \mathbf{b} A^{(j+1)} \cdots A^{(n)}] \\ =& \begin{bmatrix} a_{11} & \cdots & a_{1(j-1)} & b_1 & a_{1(j+1)} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2(j-1)} & b_2 & a_{2(j+1)} & \cdots & a_{nn} \\ \vdots & \dots & \vdots & \vdots & \vdots & \dots & \vdots \\ a_{n1} & \cdots & a_{n(j-1)} & b_n & a_{n(j+1)} & \cdots & a_{nn} \end{bmatrix} \end{aligned}$$

Theorem (Cramer's Rule)

That is,

$$
\begin{aligned}
det(A_j) =& b_1 C_{1j} + b_2 C_{2j} + \cdots + b_n C_{nj} \\
\implies x_j =& \frac{det(A_j)}{det(A)}, j = 1, 2, \cdots, n
\end{aligned}
$$

Cramer's Rule.

$$
\begin{aligned}
A\mathbf{X} &= \mathbf{b}, \quad (\det(A_j) \neq 0)\\
\implies x_j &= A^{-1}\mathbf{b} = \frac{1}{\det(A)}\operatorname{adj}(A)\mathbf{b}\\
&= \frac{1}{\det(A)}\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1}\\ C_{12} & C_{22} & \cdots & C_{n2}\\ \vdots & \vdots & \ddots & \vdots\\ C_{1n} & C_{2n} & \cdots & C_{nn}\end{bmatrix}\begin{bmatrix} b_1\\ b_2\\ \vdots\\ b_n\end{bmatrix} = \frac{1}{\det(A)}\begin{bmatrix} b_1 C_{11} & + & b_2 C_{21} & + & \cdots & + & b_n C_{n} \\ b_1 C_{12} & + & b_2 C_{22} & + & \cdots & + & b_n C_{n} \\ & & \vdots & & & & \\ b_1 C_{1n} & + & b_2 C_{2n} & + & \cdots & + & b_n C_{n} \end{bmatrix}
\end{aligned}
$$

$\square$

Cramer's Rule.

$$\implies \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} b_1 C_{11} & + & b_2 C_{21} & + & \cdots & + & b_n C_{n1} \\ b_1 C_{12} & + & b_2 C_{22} & + & \cdots & + & b_n C_{n2} \\ & & \vdots & & & & \\ b_1 C_{1n} & + & b_2 C_{2n} & + & \cdots & + & b_n C_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} \det(A_1/\det(A) \\ \det(A_2/\det(A) \\ \vdots \\ \det(A_n/\det(A) \end{bmatrix}$$

$$\implies x_j = \frac{\det(A_j)}{\det(A)}, j = 1, 2, \cdots, n$$

$\square$

Example

Use Cramer's rule to solve the system of linear equation

$$\begin{array}{rcrcrcr} -x & + & 2y & - & 3z & = & 1 \\ 2x & & & + & z & = & 0 \\ 3x & - & 4y & + & 4z & = & 2 \end{array}$$

Solution

Use Cramer's rule to solve the system of linear equation

$$det(A) = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{vmatrix} = 10$$

$$det(A_1) = \begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 2 & -4 & 4 \end{vmatrix} = 8,\ det(A_2) = \begin{vmatrix} -1 & 1 & -3 \\ 2 & 0 & 1 \\ 3 & 2 & 4 \end{vmatrix} = -15,\ det(A_3) = \begin{vmatrix} -1 & 2 & 1 \\ 2 & 0 & 0 \\ 3 & -4 & 2 \end{vmatrix} = -16$$

Therefore

$$x = \frac{det(A_1)}{det(A)} = \frac{4}{5},\ y = \frac{det(A_2)}{det(A)} = \frac{-3}{2}\ z = \frac{det(A_3)}{det(A)} = \frac{-8}{5}$$

Determinants as Area or Volume

- If A is a 2×2 matrix, the *area* of the parallelogram determined by the columns of A is $\det(A)$.

Determinants as Area or Volume

- If A is a 2×2 matrix, the *area* of the parallelogram determined by the columns of A is $\det(A)$.
- If A is a 3×3 matrix, the *volume* of the parallelepiped determined by the columns of A is $\det(A)$.

Exercise1

Use row operations and the properties of the determinant to calculate the three by three "Vandermonde determinant"

$$\det(A) = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

Exercise2

Using determinant rules, calculate determinant of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 4 & 4 & 4 \\ 1 & 4 & 9 & 9 \\ 1 & 4 & 9 & 16 \end{bmatrix}$$

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Exercise3

True or false, with reason if true and counterexample if false:

1. If A and B are identical except in the upper-left corner, where $b_{11} = 2a_{11}$, then $\det(B) = 2\det(A)$.
2. The determinant of a matrix is the product of the pivots.
3. If A is invertible and B is singular, then $A + B$ is invertible.
4. If A is invertible and B is singular, then AB is singular.